

ENR 1.5 HOLDING, APPROACH AND DEPARTURE PROCEDURES

1.5.1 General

1. The holding, approach and departure procedures in use within the Manila FIR generally conform with those contained in ICAO Doc 8168-OPS/611.
2. In the formulation of the holding areas and its attendant procedures, the following factors and values were taken into account:
 - a. Intangible Values
 - i. Pilot proficiency;
 - ii. Width of area of ambiguity at heights above marker beacons and other radio facilities;
 - iii. Effects of turbulence;
 - iv. Corrections by pilots for wind effects;
 - v. Other factors such as variations between aircraft characteristics, pilot fatigue, type of weather, etc.
 - b. Tangible Values
 - i. Maximum true airspeed of 150 knots;
 - ii. Minimum true airspeed of 90 knots;
 - iii. Wind speed of 60 knots from any direction (omni-directional). The effect of wind velocity was tempered having regard to the phase of procedure;
 - iv. Tolerances for ground and airborne equipment.

Note: Holding areas and procedures are graphically illustrated in the appropriate instrument approach charts.
3. In the instrument approach charts, the data used are as follows:
 - a. Bearings – Magnetic
 - b. Distance (longitudinal) - Nautical Miles
 - c. Distance (vertical) – Feet
 - d. Rate of descent - 500 Feet/Minute
 - e. Rate of turn - 3 Degrees/Second
 - f. Visibility - Kilometers

1.5.2 Arriving flights

1. IFR flights entering, and landing within, a Terminal Control Area will be cleared to a specified holding point and instructed to contact Approach Control at a specified time, level or position. The terms of this clearance shall be adhered to until further instructions are received from Approach Control. If the clearance limit is reached before further instructions have been received, holding procedure shall be carried out at the level last authorized.
2. IFR flights destined to locations where standard terminal arrival routes (STAR) have been established may be issued a clearance containing a STAR.
3. Due to the limited airspace available, it is of importance that the approaches to the patterns and the holding procedures are carried out as exactly as possible. Pilots are strongly requested to inform ATC if for any reason the approach and/or holding cannot be performed as required.

4. Speed control and radar sequencing (where radar service is available) may be expected from ATC in order to achieve maximum airspace utilization within the parameters of the established operational and separation minima.

1.5.3 Departing flights

1. IFR flights departing from controlled aerodromes will receive initial ATC clearance from the local Aerodrome Control Tower. The clearance limit will normally be the aerodrome of destination.
2. IFR flights departing from non-controlled aerodromes will not take off without prior arrangements with the Area Control Center concerned.
3. IFR flights departing from aerodromes where standard instrument departure (SID) or radar departure (RDR) procedures are effective may expect, in addition to the normal instructions, clearance containing a SID or RDR.
4. Departing IFR flights are required to establish on the centerline of the appropriate ATS route, or the departure radial specified in the ATC clearance, as soon as practicable, after the aircraft has reached the minimum safe altitude for terrain clearance unless following a specific SID or RDR procedure.

1.5.4 Other relevant Information and Procedures

1. Holding

1.1 Holding Clearance

- 1.1.1 A holding clearance issued by ATC normally includes the following information:
 - a. Clearance to the holding fix;
 - b. Direction to hold from the holding fix;
 - c. On specified radial, course inbound track;
 - d. If DME is used, the DME distances at which the fix end and the outbound end turns are to commence, e.g. "HOLD BETWEEN (number of miles) AND (number of miles)"
 - e. Time to expect further clearance, approach clearance, or time to leave the fix in the event of communication failure.

1.2 Indicated Airspeeds

- 1.2.1 Holding patterns shall be entered at or below the following indicated airspeeds:

Levels	Propeller Aircraft **	Jet Aircraft	
		Normal Conditions	Turbulence Conditions
Up to and including 6000FT (1850M)	170 knots	210 knots	280 knots or 0.8 Mach which ever is less ***
Above 6000FT (1850M) up to including 14000FT (4250M)	170 knots	220 knots	
Above 14000FT (4250M)	185 knots	240 knots	

* The levels tabulated above represent 'altitudes' or corresponding 'flight levels' depending upon the altimeter setting in use.

** Certain types of propeller aircraft may need to hold at higher speeds.

*** The speed of 280 knots (0.8 Mach) reserved for turbulence conditions shall be used for holding after prior clearance from ATC, unless the relevant publications indicate that the holding area can accommodate aircraft flying at these high holding speeds.

1.3 Bank/Rate of Turn

- 1.3.1 All turns are to be made at a bank angle of 25 degrees or at a rate of 3 degrees/second, whichever requires the lesser bank.

1.4 Direction

- 1.4.1 Unless the clearance given by ATC contains instruction to the contrary, all turns after initial entry into the holding pattern shall be made to the RIGHT.

1.5 Timing

- 1.5.1 Outbound timing begins over or abeam the fix whichever occurs later. If the abeam position cannot be determined, start timing when turn to outbound is completed.

1.6 Wind Effects

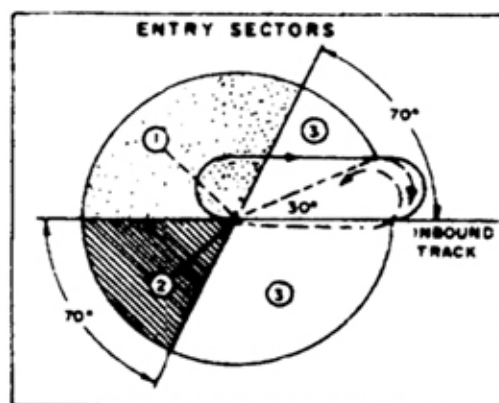
- 1.6.1 The pilot shall make allowance for known wind by applying corrections both to heading and timing during entry and while flying in the holding pattern.

1.7 Non-Conformity

- 1.7.1 If for any reason a pilot is unable to conform to the procedures for normal conditions laid down for any particular holding pattern he should advise ATC as soon as possible.

1.8 Entry Procedures (Standard Holding Pattern)

- 1.8.1 The entry into the holding pattern shall be on a heading in relation to the three (3) entry sectors shown below, recognizing a zone of flexibility of 5 degrees on either side of the sector boundaries. In case of holding on VOR intersections or VOR/DME fixes, entries will be limited to the radials or DME arcs forming the fix, as appropriate.



1.8.2 Sector 1 Procedure (Parallel Entry):

- Having reached the fix, turn onto an outbound heading for the appropriate period of time.
- Turn left to intercept the inbound track to return to the fix.
- On second arrival over the fix, turn right and follow the holding pattern.

1.8.3 Sector 2 Procedure (Offset Entry):

- Having reached the fix, turn onto a heading to make good a track making an angle of 30 degrees or less from the inbound track on the holding side.

- b. Continue for the appropriate period of time.
- c. Turn right to intercept the inbound track and follow the holding pattern.

1.8.4 Sector 3 Procedure (Direct Entry). Having reached the fix, turn right and follow the holding pattern.

1.8.5 The still airtime for flying the outbound entry heading should not exceed one minute if at or below 4250M (14000FT) or one and one half minutes if above 4250M. Where DME is available, the length of the outbound leg may be specified in terms of the distance instead of time.

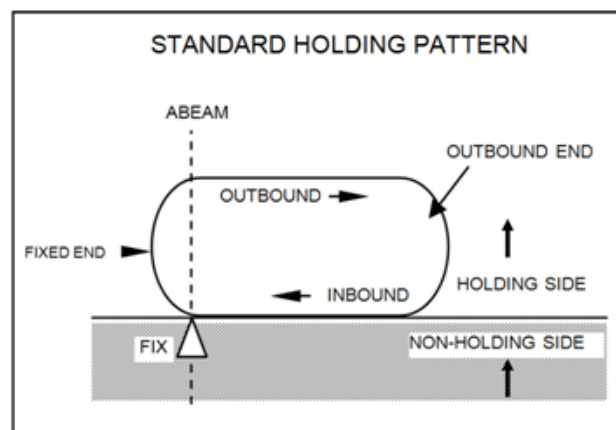
1.9 Standard Holding Pattern

1.9.1 The standard holding pattern described and depicted below is in terms of still air condition.

- a. Having entered the holding pattern, on the second and subsequent arrivals over the fix execute a turn to fly an outbound track, which will most appropriately position the aircraft for the turn onto the inbound track.
- b. Continue outbound for one minute if at or below 4250M (14000FT) or for one and a half minutes if above 4250M.

Note: Distance may be specified instead of time where DME is available.

- c. Turn so as to realign the aircraft on the inbound track.



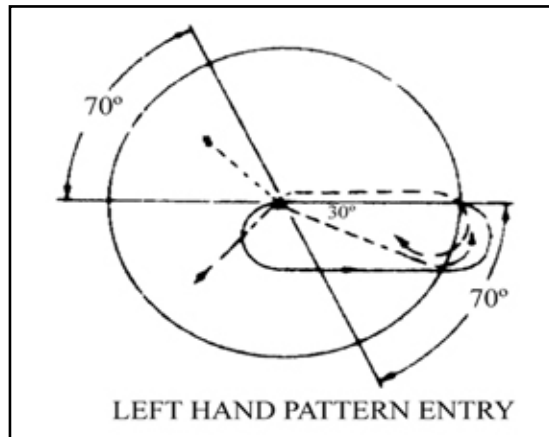
1.9.2 Due allowance should be made in both heading and timing to compensate for the effects of known wind.

1.10 Non-Standard Holding Pattern

1.10.1 A non-standard holding pattern requires that:

- a. Fix end and outbound end turns be made to the left and/or
- b. Time along the outbound track be other than the one or one and a half minute leg appropriate for the altitude being flown.

- 1.10.2 Entry procedures to a non-standard pattern requiring left turns are oriented in relation to the 70 degree line on the holding side, just as in the standard pattern (see diagram).



1.11 Departing the Pattern

- 1.11.1 When ATC clearance is received specifying the time of departure from the holding point, the pilot should adjust his pattern within the limits of the established holding procedure in order to leave the holding point at the time specified.

2. Deviation from the Holding, Approach and Departure Procedure

- 2.1 Except in an emergency, which necessitates immediate action, aircraft may not deviate from the standard holding, approach or departure procedure.
- 2.2 Aircraft that have deviated from such procedures shall inform the appropriate ATC unit as soon as possible to enable that unit to resolve any traffic confliction resulting from that action.
- 2.3 Aircraft which has been assigned a specific approach procedure, when visual reference has been established, may request for clearance to change the flight path, otherwise, the entire procedure has to be complied with.